

# Physical Chemistry (I)

## Lecture 24

기말고사 안내 및 복습



# 기말고사 안내

- 시간: 6월 12일(목) 오전 10:30~11:30
- 장소: 화학관 315호(홀수 학번), 112호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

# 시험 범위

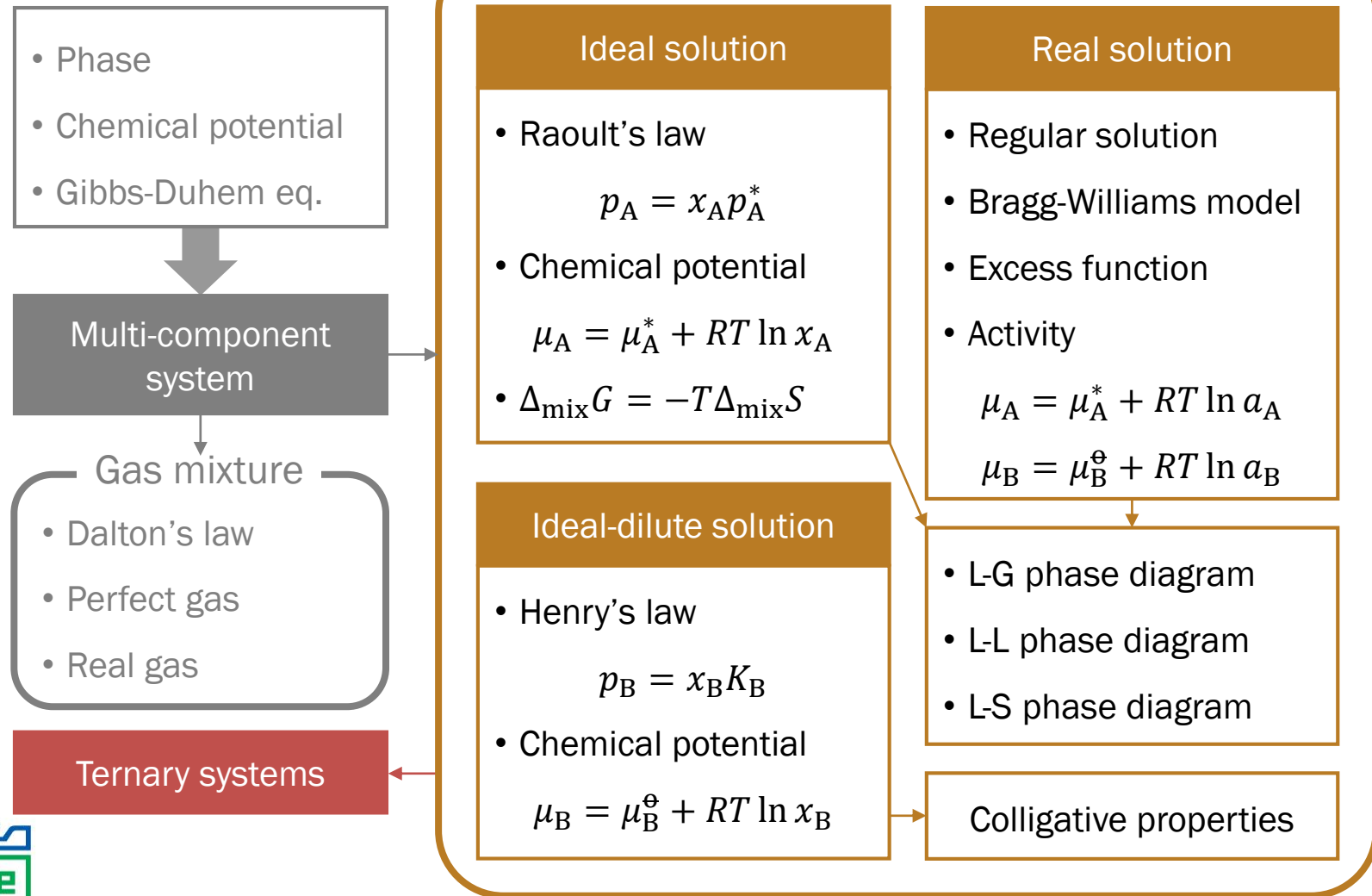
- 8장 “단순 혼합물” 중 2차 중간고사 이후 범위
- 9장 “화학 평형”
- 10장 “전기화학”

# 시험 문제

## 총 여섯 문제

- 과제 문제 혹은 그 변형(3문제)
  - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(2문제)
  - 강의 자료 기준
- 보너스 문제(1문제)

## Chapter 8



# Concepts and Derivations in Chpt 8

- [C] Mixing of liquids: liquid-gas equilibrium
- [C] Ideal solutions
  - [C] Raoult's law and ideal solutions
  - [D] Chemical potential
  - [D] Thermodynamics of mixing
- [C] Ideal-dilute solutions
  - [C] Henry's law and ideal-dilute solutions
  - [C] (Idealized) chemical potentials of solute and solvent

# Concepts and Derivations in Chpt 8

- [C] Regular solutions
- [C] Bragg-Williams model
  - [D] Entropy of mixing
  - [D] Energy of mixing
  - [D] Helmholtz energy of mixing
- [C, D] Excess function
- [C, D] Activity and activity coefficient
  - [D] Activity of Bragg-Williams model

# Concepts and Derivations in Chpt 8

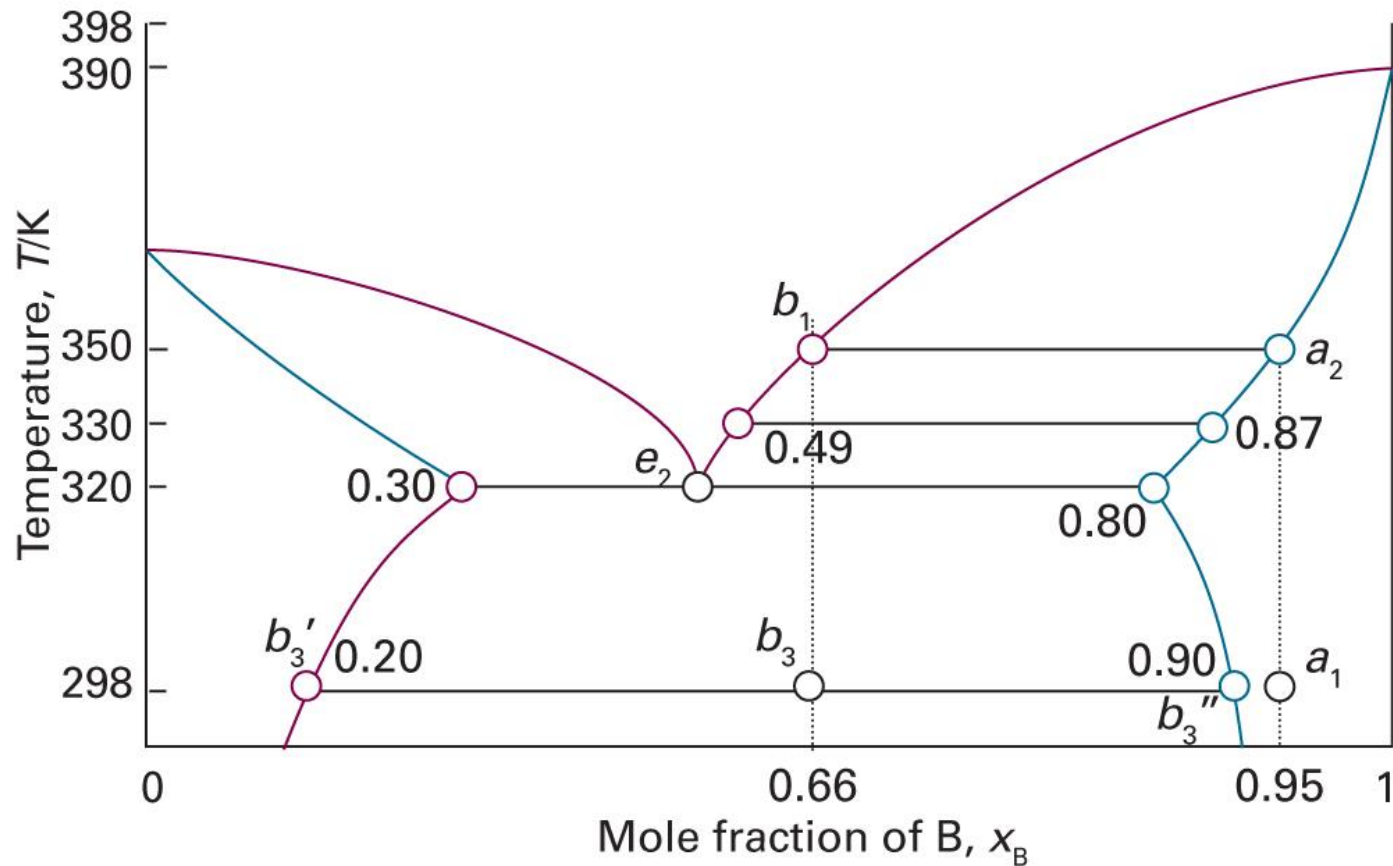
- [C, D] Phase diagrams of ideal solutions
  - [D] Liquid and vapor compositions
  - [C] Interpretation of phase diagrams
  - [C, D] Lever rule
- [C, D] Phase diagrams of real solutions
  - [C] azeotropes
  - [C, D] phase separation
- [C] Phase diagrams of liquid-solid mixtures



# Concepts and Derivations in Chpt 8

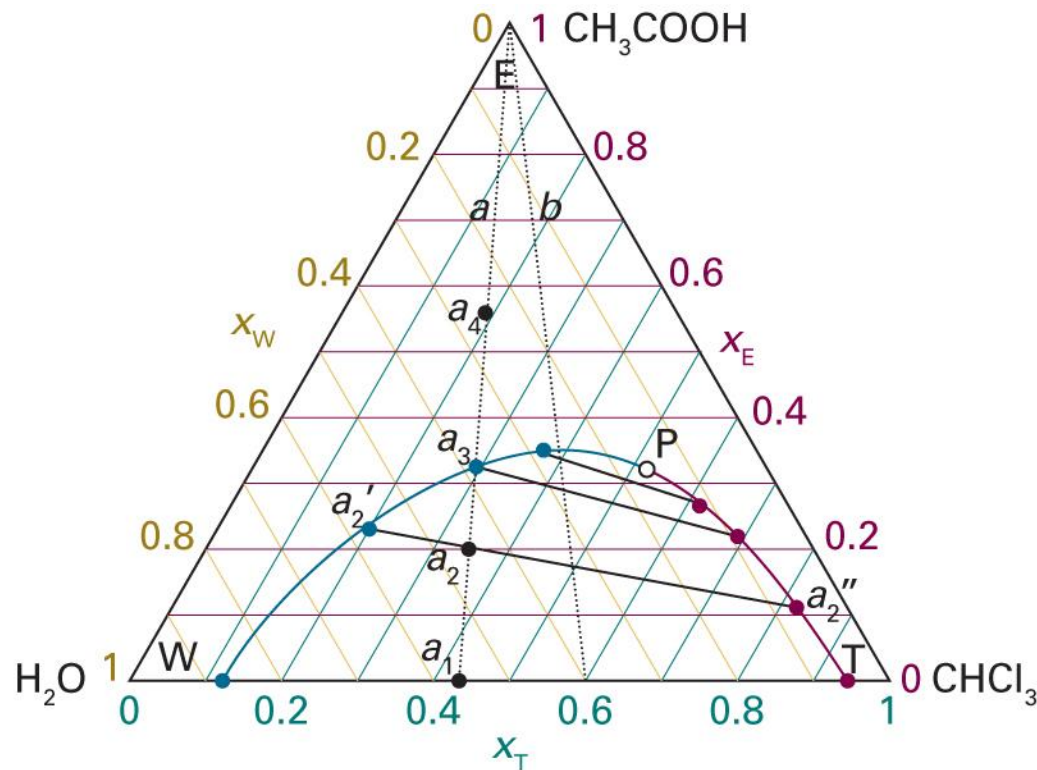
- [C, D] Colligative properties
  - Lowering of vapor pressure
  - Elevation of  $T_b$  and depression of  $T_f$
  - Osmotic pressure
  - van't Hoff factor
- [C] Phase diagrams of ternary systems

# Interpretation of a Phase Diagram



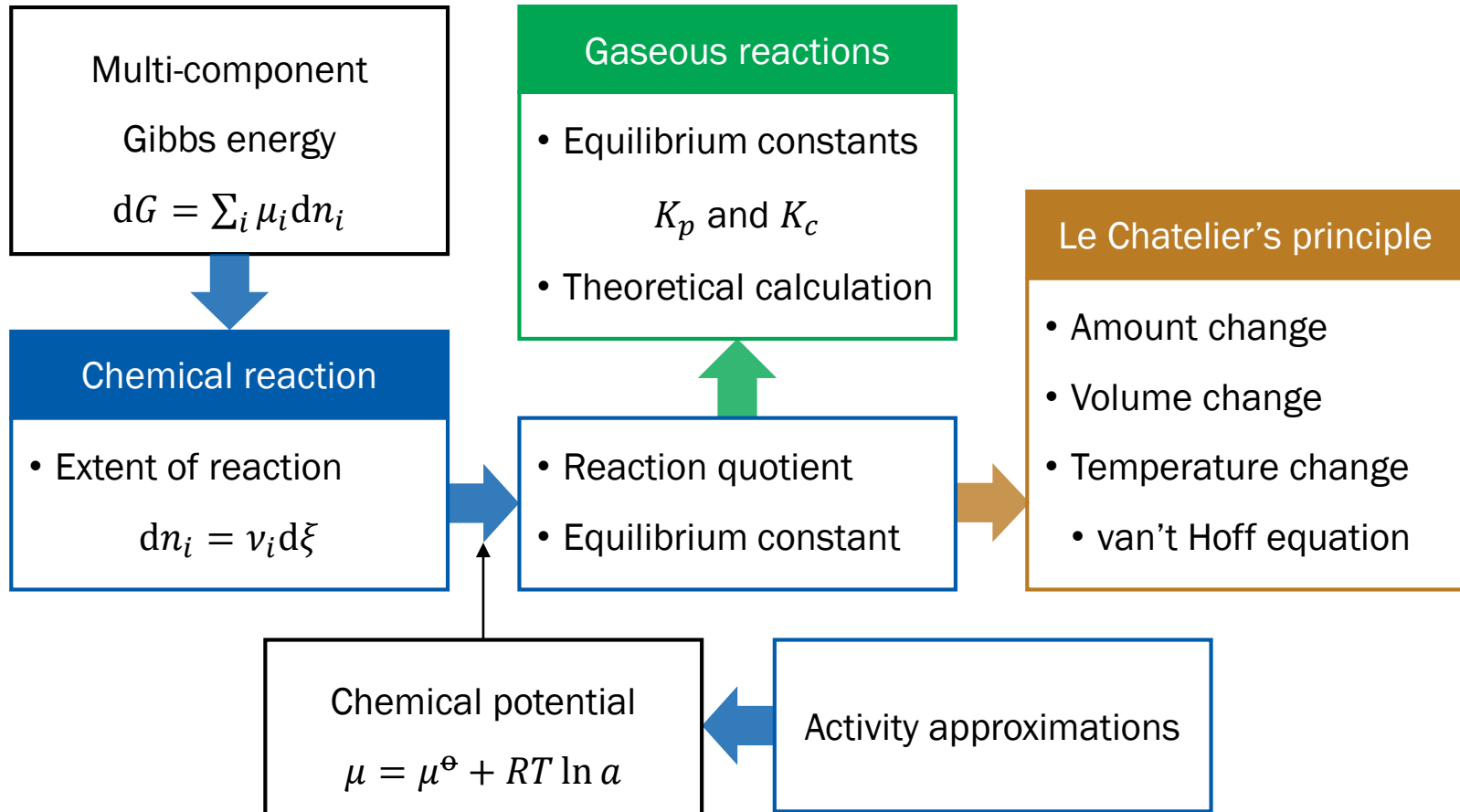
(Image: Atkins et al., *Atkins' Physical Chemistry*, 12<sup>th</sup> ed., Figure 5C.23)

# Interpretation of a Phase Diagram



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12<sup>th</sup> ed., Figure 5E.4)

## Chapter 9

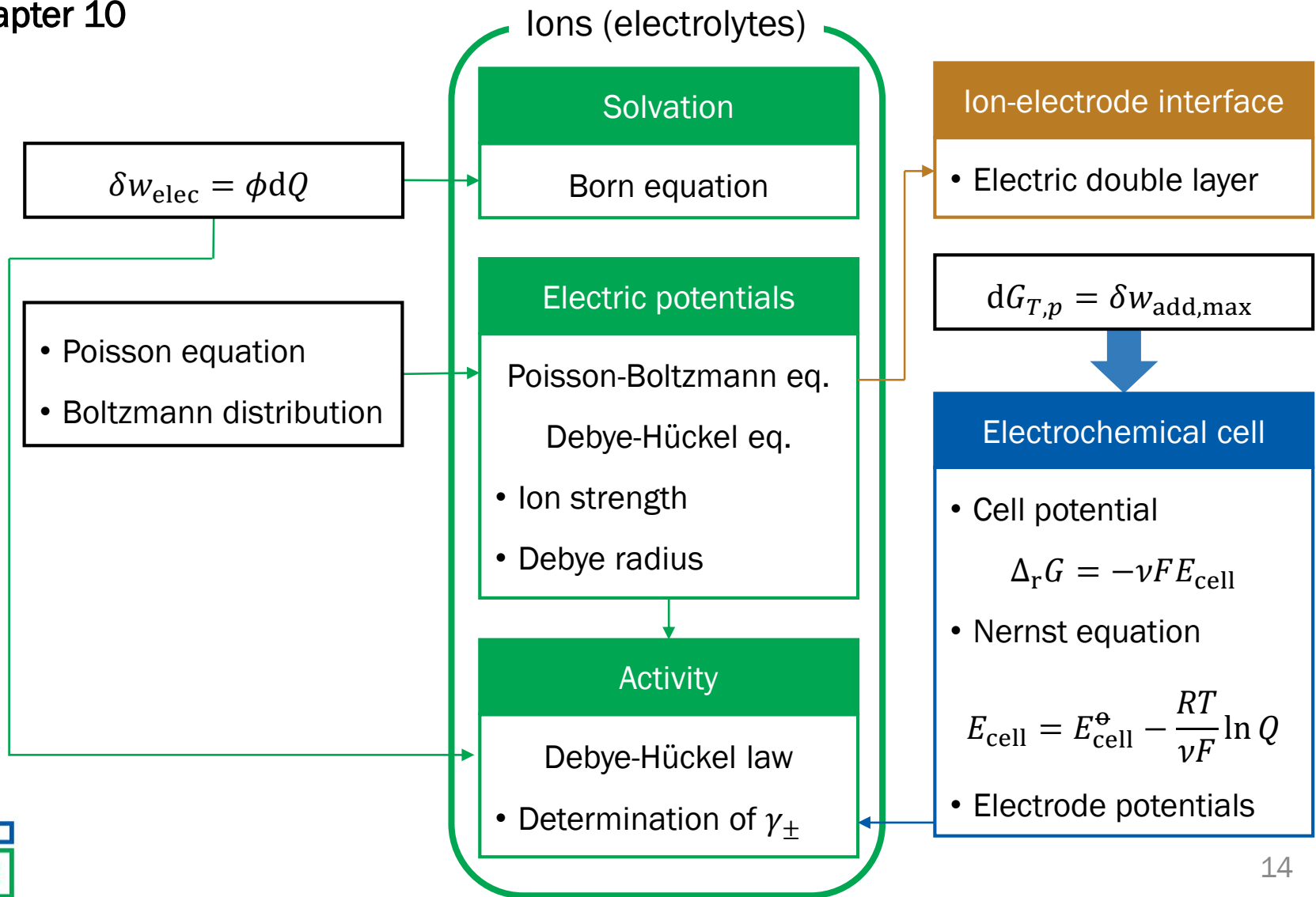


# Concepts and Derivations in Chpt 9

- [C, D] Reaction quotient and equilibrium constant
- [C, D] Activity approximations
- [D] Gaseous reactions
  - [D]  $K_p$  and  $K_c$
  - [D] Calculation of  $K_c$  from partition functions
- [C, D] Le Chatelier's principle
  - Amount change
  - Volume change
  - Temperature change: van't Hoff equation



## Chapter 10



# Concepts and Derivations in Chpt 10

- [C, D] Thermodynamics of non-expansion works
- [C] Electrochemical cells
- [D] Thermodynamics of cells
- [D] Nernst equation
- [D] Entropy and enthalpy of cell reactions
- [C, D] Electrode potentials

# Concepts and Derivations in Chpt 10

- [C, D] Gibbs energy of ion solvation: Born equation
- [D] Ion potential: Debye-Hückel equation
- [D] Ion activity: Debye-Hückel limiting law
- [D] Harned cell: determination of  $E^\ominus$  and  $\gamma_{\pm}$
- [C] Electrode-ion interface: electric double layer